

Whole-Home Back-up solution

Via *External ATS*



Storage&Battery

Introduction

In general, there are two mainstream solutions for household loads distribution on the market.

1. **Partial-Home Back-up** solution

In this solution, users' household loads will be separated into **Common Loads** and **EPS Loads**, only **EPS Loads** can be powered during **EPS mode**.

2. **Whole-Home Back-up** solution

In this solution, users' whole household loads will be able to be powered during **EPS mode** if power supply is enough.

These two solutions are well related to the transferring methods between **On-grid** & **EPS** status.

External ATS (Automatic Transfer Switching Device) is an equipment installed outside of the inverter, which can automatically transfer between **On-grid** & **EPS** status. That is to say, if users want to power their whole household loads during off-grid, an **External ATS** device is necessary [for G4 inverter](#). There are three products provided by SolaX that serve as **External ATS**:

1. **EPS Box**: With a **DPDT** equipment contained;
2. **MateBox Basic & Advanced**: An integration of **DPDT** and other accessories;
3. **EPS Parallel Box**: A special **EPS Box** for EPS parallel solution.

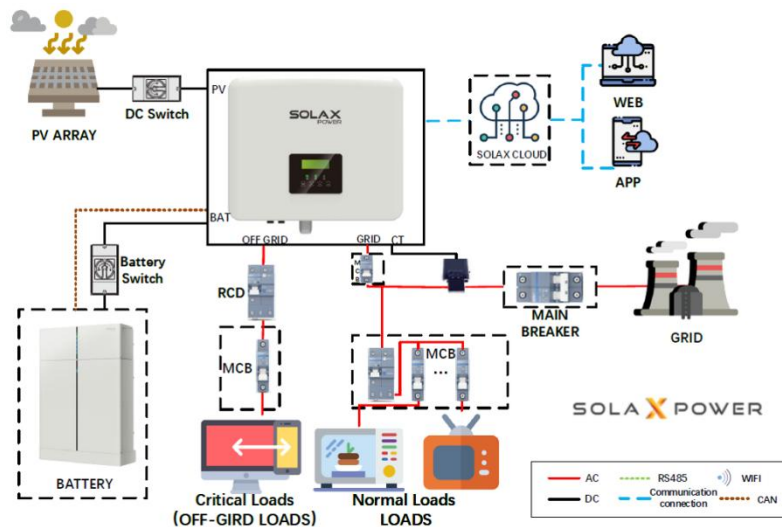
Moreover, [**External ATS**] also means a commission on the LCD settings of inverters.

Partial-Home Back-up & Whole-Home Back-up

1. **Partial-Home Back-up**

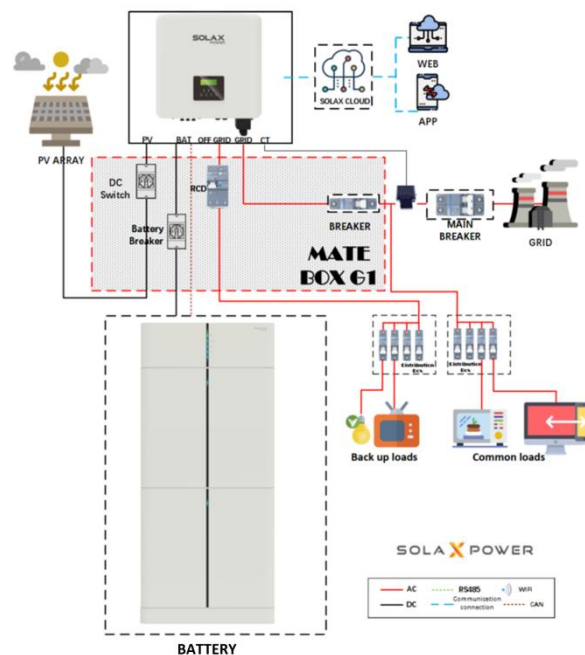
As for **Partial-Home Back-up**, built-in transferring switches will be used to change the output port when the inverter enter into EPS mode causing by power outage. And the users' household loads have to be separated into **Common Loads** and **EPS Loads**. Only **EPS Loads** will be powered during **EPS** mode such as lights and refrigerators . The system diagram is shown below.

① Normal system



Note: Need to disable **[External ATS]**, without MateBox (need to install external protection devices).

② With Matebox Basic



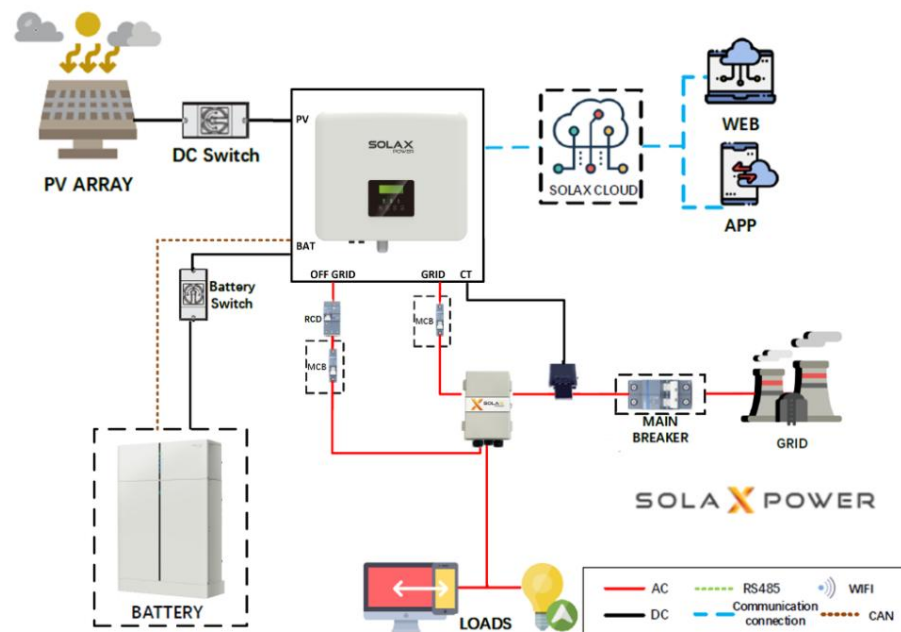
Note: Need to disable **[External ATS]**, with Matebox Basic, with T30 Batteries to form All-in-one Solution.

2. Whole-Home Back-up

Whole-Home Back-up can be achieved by external switching devices for **Hybrid inverters** and the inverters should have the **External ATS** function, in which users'

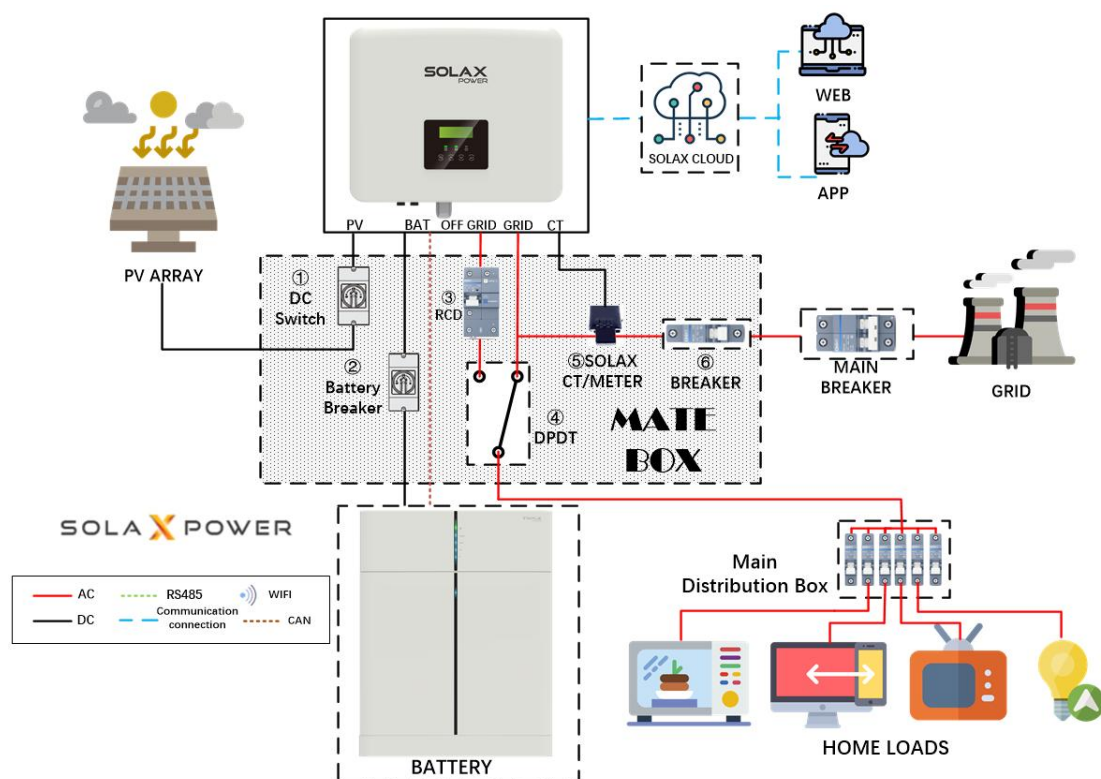
whole household loads can be powered during off-grid. A **Whole-Home Back-up** solution is shown below.

① With EPS Box



Note: Need to Enable [External ATS], the load power should be no more than the maximum EPS output power.

② With Matebox Advanced



Notes:

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1. Need to Enable [External ATS]
 2. The total power of the loads run simultaneously should be no more than EPS output power.
 3. Different transferring devices cannot be applied simultaneously, for example, a MateBox Advanced shall not exist in parallel solutions with EPS Parallel Box.
 4. For X3-Ultra, the whole home loads can be connected to EPS port to realize the whole-home backup by the internal transferring switch.

Advantages

1. **EPS Box:**

Cost-saving

EPS Box is much cheaper than **MateBox Advanced** is. For those inverters that have been equipped with Meter/CT or other accessories, an **EPS Box** is prefer.

Notes: EPS Box has been eliminated gradually.

2. **MateBox Basic & Advanced:**

Easy-to-install

Thanks to the pre-wired solution, **MateBox** will assemble all accessories and cables in advance, omitting complicated installation steps.

Eco-friendly

Simple installation steps can save labor costs.

Clean & Tidy

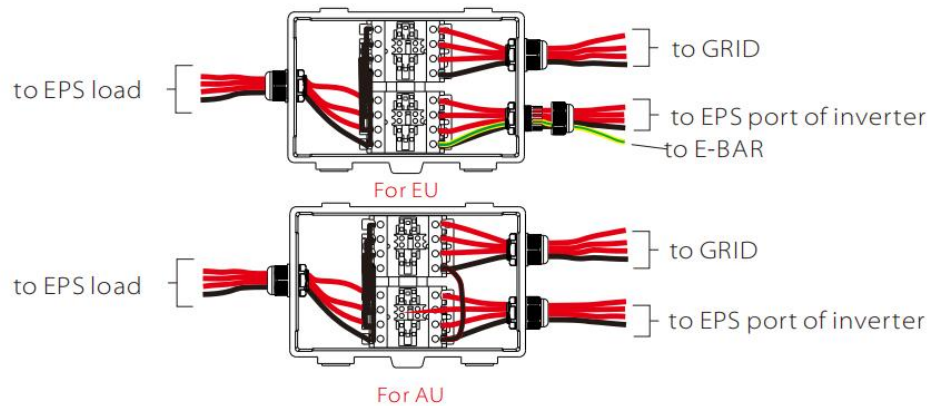
All wirings are operated inside the box and cabinet, and the outside of the entire system will be clean and tidy.

Configuration

Cable Connection:

1. X3-EPS Box:

Please make sure all wires are tightened. Wire connection in Three-phase EPS-Box.



2. X3-Matebox Basic

Diagram A: N line and PE line separate wiring, D series inverters; (For most countries)

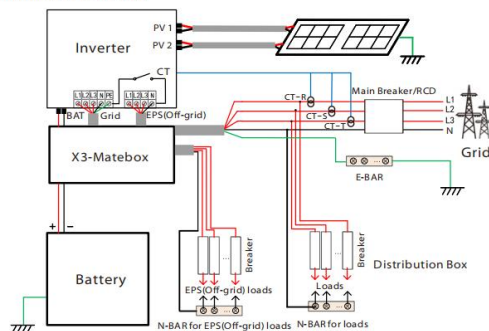
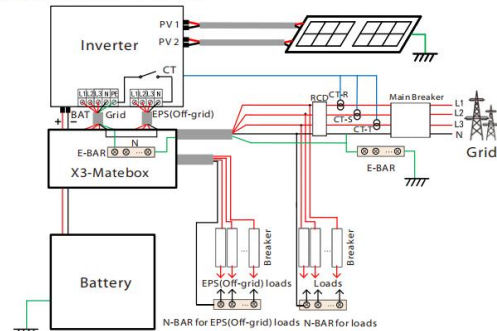


Diagram B: N line and PE line together, D series inverters; (Applicable to Australia)



3. X3-Matebox Advanced:

Diagram A: N line and PE line separate wiring, M series inverters; (For most countries)

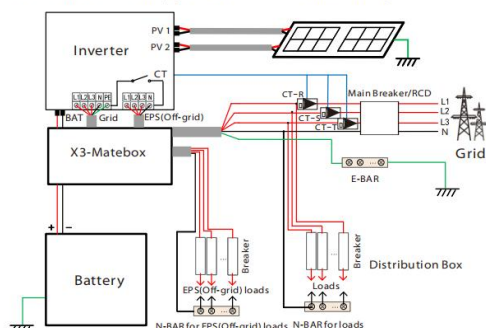
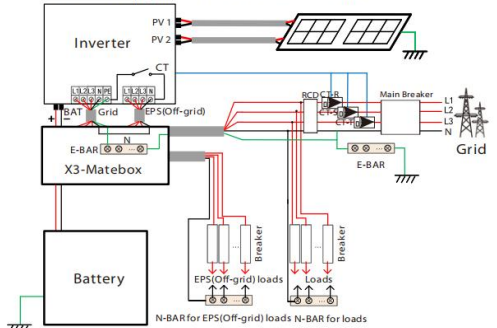
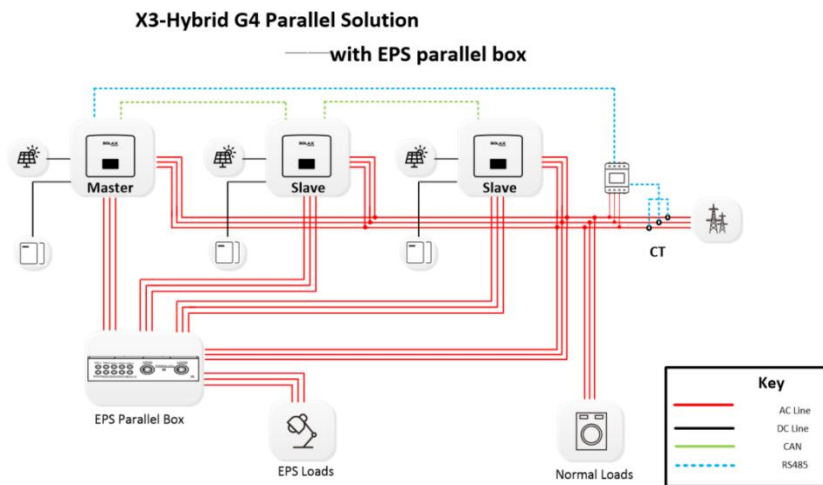


Diagram B: N line and PE line together, M series inverters; (Applicable to Australia)





Settings & Commissions:

Setting path:

Main → Settings → Advance Setting → External ATS

This setting is **Disable** by default. If any **External ATS** device (EPS Box, EPS Parallel Box, MateBox Advanced) mentioned above is used, select **Enable** to enable this function.

Compatibility

[External ATS] Hardware version:

X1-Hybrid/Fit G4

X1-Hybrid-N/D-E G3(without setting on LCD)

X3-Hybrid-N/D-E G2(without setting on LCD)

Matebox:

X1-Matebox V1.3 + X1-Hybrid/Fit-M G4

X1-Matebox V1.4 + X1-Hybrid/Fit-D G4

X3-Matebox V1.3 + X1-Hybrid/Fit-M G4

X3-Matebox V1.4 + X1-Hybrid/Fit-D G4

EPS Box:

X1-EPS Box + X1-Hybrid G3/G4

X3-EPS Box + X1-Hybrid G2/G4

EPS Parallel Box:

X3 Hybrid G2: [X3-EPS-BOX-P5-I/E](#) & [X3-EPS-P10-I/E](#)

X3 Hybrid/Fit G4:[X3-PBOX 60/150kW-G2](#)

X3 IES: X3-PBOX 60/150kW-G2

X3 Ultra: X3-PBOX 150kW/300kW-G2